RNAlater (sample preservation solution): recipe

Huang MEGA Lab (NCKU)
v1.1, 2026-10-21; sample preservation solution; hand-maintained since 2026-10-21

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Abstract

RNAlater recipe

Procedure

BEFORE YOU START

1. Submerge all your equipment and glassware in **1% bleach to get rid of nuclease**

REAGENTS

2. Use **DEPC-treated water** for every solution below (see the lab protocol *DEPC-treated water*; store the treated water at **4 °C** or room temperature, tightly closed).

3. 0.5 M Disodium dihydrate EDTA:

1. Add **18.61 g** disodium dihydrate EDTA in **100 mL** ddH2O
2. Adjust pH to 8.0 with NaOH while stiring

4. 1 M tri-Sodium Citrate Dihydrate:

Add **29.4 g** tri-Sodium Citrate Dihydrate in **100 mL** ddH2O, stir to dissolve

5. 1 M sulfuric acid:

Add **5.33 mL** sulfuric acid in **100 mL** ddH2O, stir to dissolve

START TO MAKE RNALATER

6. Make a mixture of:

40 mL of **0.5 M EDTA** + **25 mL 1 M tri-Sodium Citrate** + **700 g Ammonium Sulfate** + **935 mL ddH2O**; stir on a hot plate with low heat until ammonium sulfate dissolved

7. Allow it to cool; adjust the pH of the solution to **pH 5.2** using **1 M sulfuric acid**

Note To avoid contamination from the immersed pH meter, adjust pH by pouring 10% of the solution in a new beaker and add base or acid to hit the right pH; then scale it up to the needed amount for the original solution and blindly adjust its pH

8. The final concentration: 25 mM Sodium Citrate, 10 mM EDTA, 70 g ammonium sulfate in 100 ml ddH2O; pH 5.2.

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9. Transfer to screw top, **nuclease-free** bottles; label the date of make; store at **4 °C**